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# Leiserchess 2025

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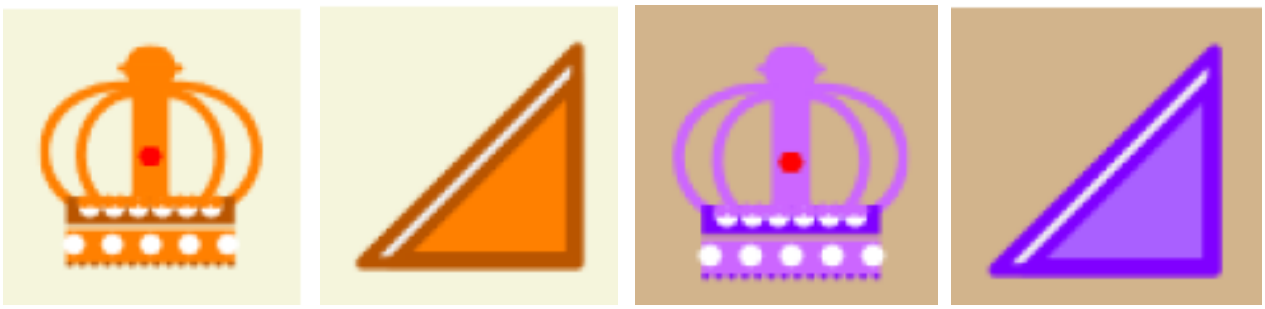
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(a) Tangerine Monarch. (b) Tangerine Pawn. (c) Lavender Monarch. (d) Lavender Pawn.

Figure 1: The Leiserchess pieces.

# 1 Introduction

Leiserchess (pronounced "LYE-sir-chess") 2025 is a two-player laser-chess game similar to Laser Chess, Khet, and previous versions of Leiserchess.

The teaching staff of the MIT class 6.106 Software Performance Engineering developed Leiserchess 2024 (henceforth just Leiserchess) for the term final project in Fall 2024. The students are given a working implementation of a Leiserchess bot, and their job is to make it run as fast as possible and otherwise improve its playing ability. We will also use Leiserchess as a fun way to illustrate various software performance engineering concepts.

On the surface, Leiserchess is much simpler than Laser Chess or Khet. There are only two kinds of pieces Monarchs and Pawns and all pieces can move only to diagonally adjacent squares. Complexity arises from the dynamics of how pieces interact, however, because the Monarchs carry lasers to shoot at each other and at each other's Pawns. The result is an entertaining and challenging game that involves both tactics and strategy.

# 2 Pieces and board

Leiserchess is played by two players, ***T*angerine** and ***L*avender**, on an 8-by-8 chessboard. Each player starts with a ***M*onarch** and eight ***P*awns**, as shown in Figure 2. The 64 squares of the board can be labeled using a chess-like algebraic notation. From Tangerine's point of view, the vertical columns of squares running from bottom to top (or south to north), called files, are labeled a through h from left to right. The horizontal rows of squares running from left to right (or west to east), called ranks, are numbered 0 to 7 from bottom to top, thereby giving each square a unique coordinate consisting of a letter and a number.

The Monarchs contain lasers that shoot at the end of a player's turn in the direction of the point of the crown. The hypotenusal side of a Pawn is a mirror oriented at a 45° angle to the ranks and files of the board. The Pawns' mirrors can deflect the path of a laser beam from rank to file and vice versa. If a laser hits a Monarch or the nonreflecting side of a Pawn, that piece is destroyed and removed from the game.

## 3 The opening position

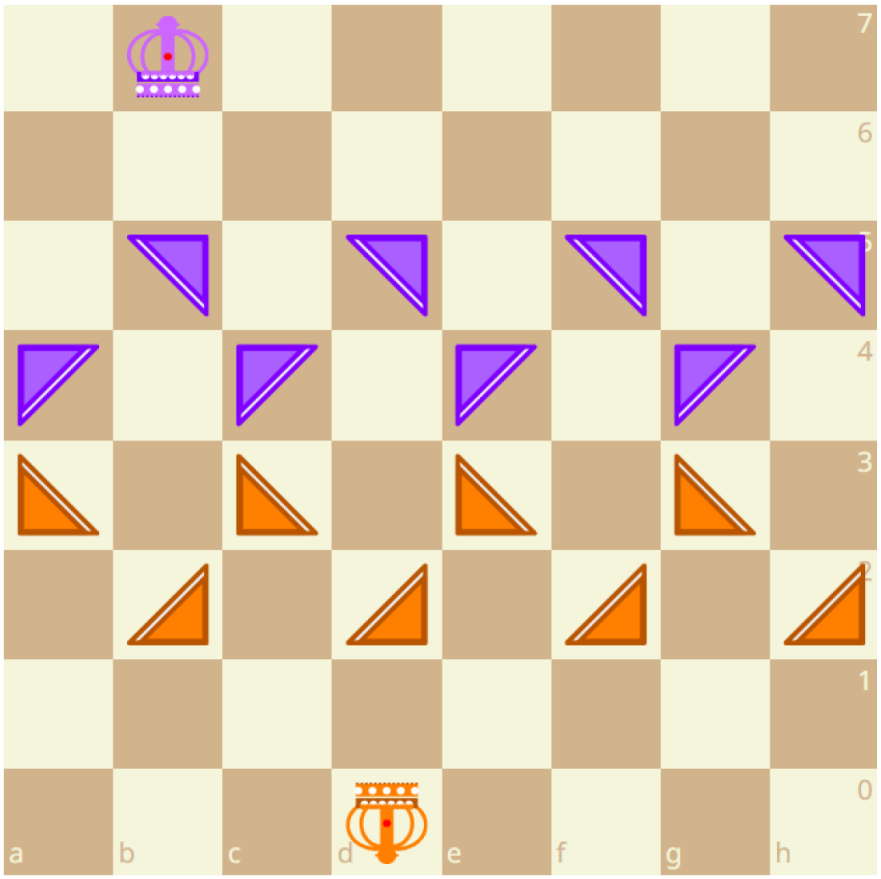



Figure 2: The opening position.

Play begins with Tangerine moving first from the position shown in Figure 2, and then play alternates. Tangerine pieces always occupy light squares on the board, and Lavender pieces always occupy dark squares. A player may move only their own pieces. A player's turn has two parts: first, the player moves a piece, and then the player fires their Monarch's laser.

## 4 Moving

For the moving part of a turn, the player on move selects one of their pieces, their Monarch or one of their Pawns, and performs one of four actions:

- **Rotate**: The player rotates the piece by 90°, 180°, or 270° clockwise. For example, from the position shown in Figure 3(a), Figures 3(b)-(d) show the three possible rotations of the Tangerine Pawn.
- **Slide**: The player slides the piece one square diagonally to an empty square in the northeast, northwest, southwest, or southeast direction. For example, from the position shown in Figure 3(a), Figures 3(e)-(h) show the four possible slides of the Tangerine Pawn.

- **Jump**: To jump, the piece must be a Pawn (not a Monarch), and there must be an adjacent opponent's Pawn (not the opponent's Monarch) to the north, west, south, or east. Additionally, there must be an empty square on the other side of the opponent's Pawn that is not adjacent to another one of the opponent's Pawns (not their Monarch). The player's Pawn "jumps" the opponent's Pawn, which is removed from the board, and takes occupancy of the empty square on the other side of the removed opponent's Pawn. For example, from the position shown in Figure 4(a), the Lavender Pawn can jump the Tangerine Pawn, resulting in the position in Figure 4(b). In the position Figure 4(c), the Lavender Pawn cannot jump the adjacent Tangerine Pawn because it would land next to another Tangerine Pawn. Double jumps, as in the game of checkers, are not allowed.
- **Null**: The player chooses not to move any piece. This null move is legal only if their Monarch's laser zaps a piece (see Section 5). That is, the board position must change after the null move.

A piece cannot move off the board, whether by sliding or jumping.

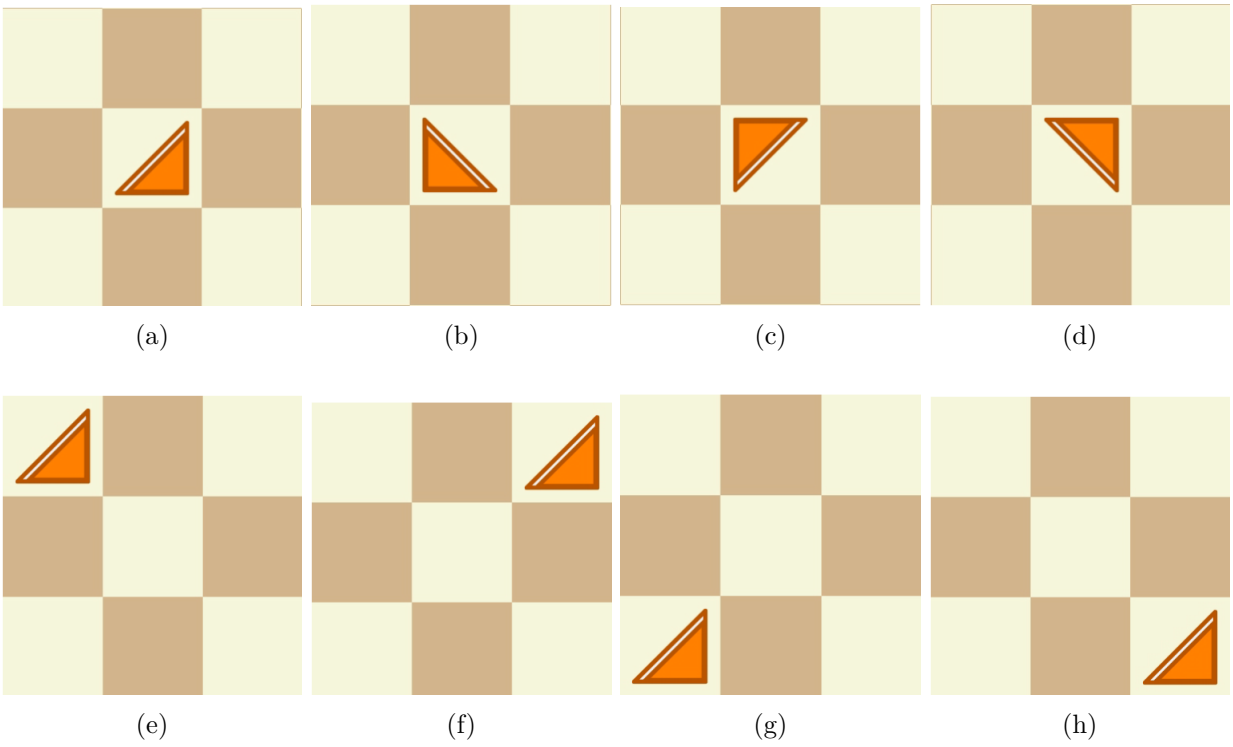
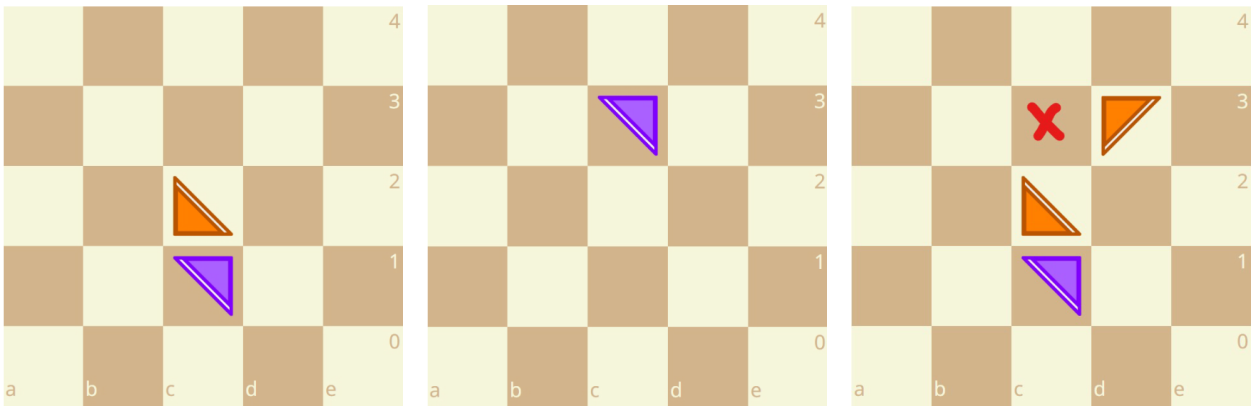



Figure 3: Possible moves for each piece. (a) The initial position of a Pawn. (b) Rotating 90°. (c) Rotating 180°. (d) Rotating 270°. (e) Sliding northwest. (f) Sliding northeast. (g) Sliding southwest. (h) Sliding southeast.

### Jumping

If a player's Pawn is adjacent to an enemy Pawn, it can jump over the enemy Pawn and capture it (remove it from the board) as long as the square on which the player's Pawn would land is not adjacent to any other enemy Pawn, as shown in Figure 4. A Pawn cannot jump off the board, and double jumps are not permitted. In the valid jump of Figure 4, the Tangerine Pawn at position c2 is captured in the process.




(a) (b) (c)

Figure 4: Jumping. (a) Initial position. (b) Valid jump by Lavender Pawn capturing Tangerine Pawn. (c) The jump is illegal because the square c3 is adjacent to another opponent Pawn on square d3.

## 5 Firing the laser

For the second part of a turn, the player on move fires their Monarch's laser, which shoots out the top of the Monarch, possibly bouncing off the mirrored surfaces of Pawns, as demonstrated in Figure 5. If the laser beam eventually passes harmlessly off the edge of the board, the turn is over. Otherwise, if the laser hits the nonreflecting side of a Pawn, the Pawn is zapped and removed from the board, and the turn is over. If the laser zaps a Monarch, the game is over, and the player whose Monarch was zapped loses. (Try not to commit suicide, and beware of creating "friendly" fire!)

## 6 End of game

When a player's Monarch is zapped, the game is over, and the player's opponent wins. The game can also end in a draw in one of three ways:

1. There have been 50 moves by each side without a piece being destroyed. (In a tournament, the referee may replace the number 50 with another number.)
2. The same position repeats itself three times with the same player on move.
3. The two players agree to a draw. (In computer play, this rule is usually ignored.)

Unlike in chess, there is no stalemate in Leiserchess. A draw cannot occur due to a player's lack of legal moves, because a legal move is always possible.

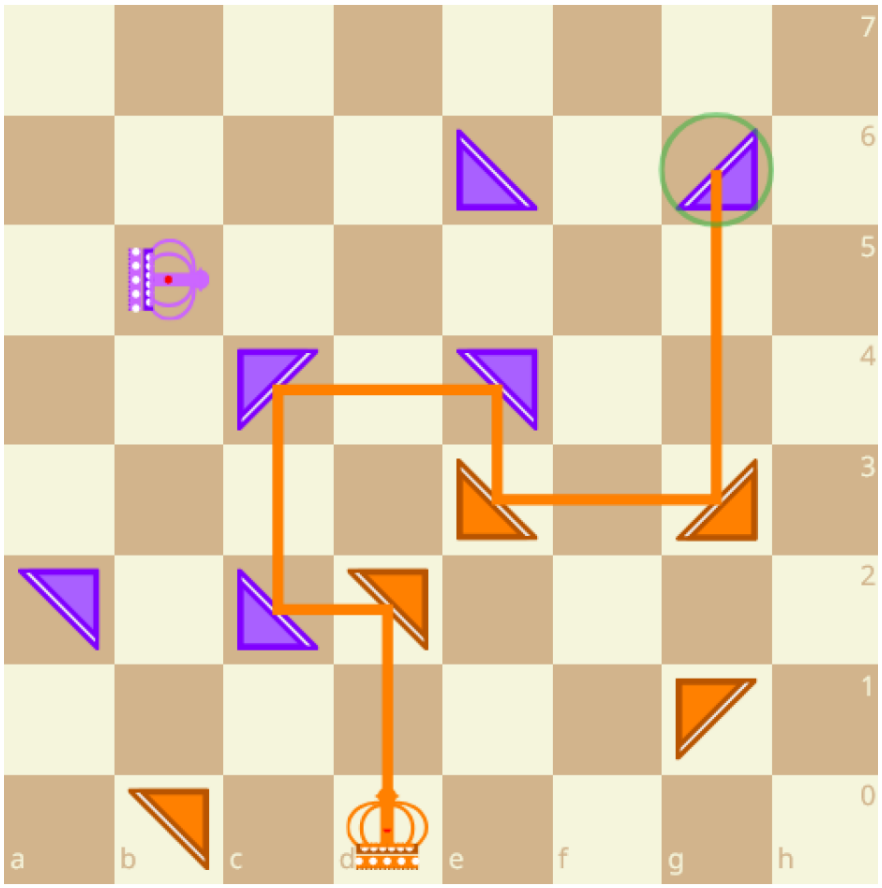



Figure 5: At the end of Tangerine's turn, Tangerine's Monarch fires its laser. The laser bounces of the mirrors of several Pawns before hitting the nonreflecting side of Lavender's Pawn at g6, which is destroyed and removed from the board.

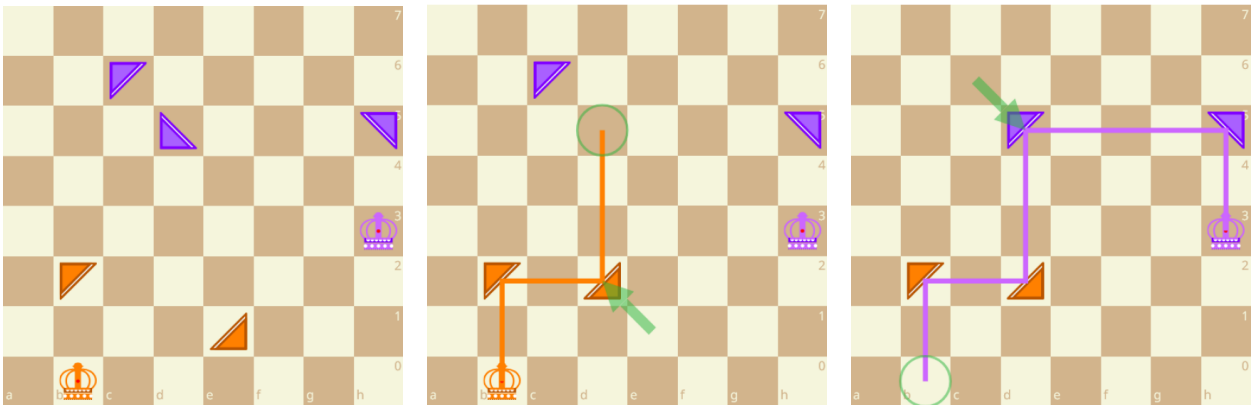
## 7 Time controls

As players become skilled, they tend to think longer. A chess clock can be used to maintain the pace. We will employ a Fischer clock, where an increment of time is added to a player's clock after their turn. Free chess-clock applications are available for many smart phones.

## 8 Learning Leiserchess

To learn the game, start by clearing the board of all Pawns and play with just the Monarchs. You will discover that one of the two players can always force the other player's Monarch to the edge of the board where it can be zapped. Playing this endgame will give you a feeling for the power of the laser-slinging Monarchs. After you've mastered Monarch-on-Monarch play, you'll find that your play in the normal game, where Pawns deflect the laser beam, improves quickly.

# 9 Tactics



(a) A position with Tangerine to move.

(b) Tangerine moves its Pawn on e1 to d2, zapping the Pawn on d5.

(c) Lavender moves c6d5, zapping the Tangerine Monarch.

Figure 6: An illustration of Leiserchess tactics.

Despite the simplicity of the rules, Leiserchess has compelling tactics. For a Monarch to zap the enemy Monarch, it risks opening itself up to counterattack, and so shots must be artfully composed. For example, consider the configuration shown in Figure 6(a). By moving its Pawn on e1 to d2, Tangerine can destroy the Lavender Pawn on d5 as described at Figure 6(b). Doing so results in immediate disaster, however, as Lavender counters by moving its pawn from c6 to d5, zapping the Tangerine Monarch, and winning the game at Figure 6(c). Pawns at b2 and d2, which deflected Tangerine's laser to zap the Lavender Pawn, are now employed by Lavender in the reverse direction to zap the Tangerine Monarch. Don't get hoisted by your own petard!

> "If you can see me, I can see you!"

This kind of tactic illustrates the emergent complexity inherent in Leiserchess, where the interaction of simple pieces engenders complex behavior. The reverse-path nature of mirrors produces a wealth of tactics to exploit and be wary of. Often, what becomes important in Leiserchess is the threat of zapping rather than the actual act of zapping. As a result, the actual area that your Monarch's laser covers may not be too important. What tends to matter is the area that could be covered if you combine a move action with a Monarch zap. Similarly, the threat of a jump can be more important than the actual occurrence of a jump.

# 10 Strategies

A player can attain a dominant position by moving their pieces strategically so that they can eventually engineer a tactical zap. Although Leiserchess 2025 is a young game, some strategic elements have begun to emerge from games played thus far:

- Try to keep some Pawns near your Monarch. A "naked" Monarch is generally easier to zap than a Monarch with nearby bodyguards. Also, try to move some Pawns near the enemy Monarch, the better to deflect your Monarch's laser to zap it. But avoid letting your opponent use your own Pawns' mirrors against you!

- Obtain a central position for your Monarch relative to your opponent's Monarch, especially as the endgame approaches. After most of the Pawns are eliminated, a centralized Monarch can more easily chase down and zap an edge-lurking opponent.
- Keep Pawns active. As the Monarchs move toward the center, a Pawn on the edge of the board can easily find itself unable to "cooperate" with other Pawns in directing the path of the laser, rendering the Pawn on the edge next to useless.
- Keep your Pawns close to each other so that they can protect each other from being jumped.
- Limit the mobility of the enemy Monarch by threatening to attack the squares near it. Make yourself ready to attack the enemy Monarch should it move to a nearby square.
- Point your Monarch's laser in the direction of your opponent's Monarch. Getting turned around makes it harder to retaliate because your Monarch may need to both rotate and slide as opposed to just one or the other.

## 11 Recording board positions

Board positions can be recorded using a modified Forsyth-Edwards notation (FEN). A FEN string is always composed from Tangerine's point of view. To create a FEN string for a position, consider the pieces rank by rank, starting with rank 7 and ending with rank 0. Within each rank, the FEN string describes the contents of each square from a to h as follows. Each piece is encoded by a two-letter sequence describing which way it is facing, where upper case letters stand for Tangerine, and lower-case letters stand for Lavender. Specifically, the Tangerine Monarch is encoded as `NN`, `EE`, `SS`, or `WW`, depending on whether it is facing north (toward the higher-numbered ranks), east (toward higher-lettered files), south, or west, respectively. The Lavender Monarch is similar, but using lower-case letters: `nn`, `ee`, `ss`, and `ww`. A Tangerine Pawn is encoded using upper case as `NE`, `SE`, `SW`, or `NW`, depending on whether its mirror is facing northeast, southeast, southwest, or northwest, respectively. Similarly, a Lavender Pawn is encoded using lower case as `ne`, `se`, `sw`, or `nw`. The numbers 1 through 8 indicate consecutive empty squares, and a forward slash separates ranks. After the board description, either the letter W or B (for White and Black, the traditional chess colors) follows, depending on whether Tangerine (W) or Lavender (B) has the next move.

For example, the following FEN string describes the opening position:

```
1nn6/8/1swlswlswlsw/selselselse1/NE1NE1NE1NE1/1NW1NW1NW1NW/8/3SS4 W
```

Optional Exercise: Write the FEN string for each of the positions in Figure 6.

## 12 Recording games

Games can be recorded using the following simple algebraic notation:

1. If a piece is rotated, write down the square containing the piece followed by
   - L for a counterclockwise rotation
   - R for a clockwise rotation e.g., `g2R`
   - U for a 180° rotatione.g., `g2U`

   e.g., `g2L`

2. If a piece is moved to an adjacent square, write down the source square followed by the destination square e.g., `h3g4`

3. To record the null move, move the Monarch to its own square e.g., `a2a2`. (Remember that a null move is illegal unless something is zapped.)

4. Record the outcome of the game as 1-0 (Tangerine wins), 0-1 (Lavender wins), or 1/2-1/2 (draw).

Here's an example of a recorded game:

```
1. c3R b5U          8. e3e1 h5g6        15. h2h4 a6R
2. d2L c4d3         9. f6h6 a4b3        16. c3L d5R
3. d0c1 670         10. c3a3 c6d5       17. c1L d5L
4. a3b4 d5c6        11. b2c2 b7a6       18. c3b2 d5c4
5. c1L f5R          12. h6U g4h3        19. c1b0 c4d3
6. g3f4 e4f3        13. b4b6 a6L        20. b0a1
7. f4f6 f3e2        14. h6g5 h3R
```

1-0

Optional Exercise: Use algebraic notation to record the sequence of moves in Figure 6.

Optional Exercise: Play Leiserchess with a friend or with the Leiserchess-playing program, and record the moves. Then replay the game from your record.

## 13 Feedback

Please post questions or comments to Piazza. Thank you, and we hope you enjoy Leiserchess 2025!